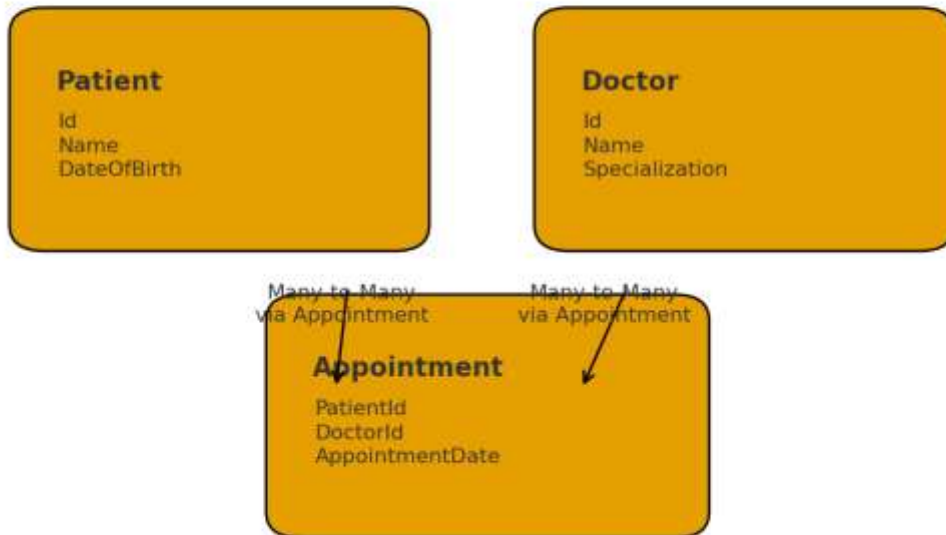
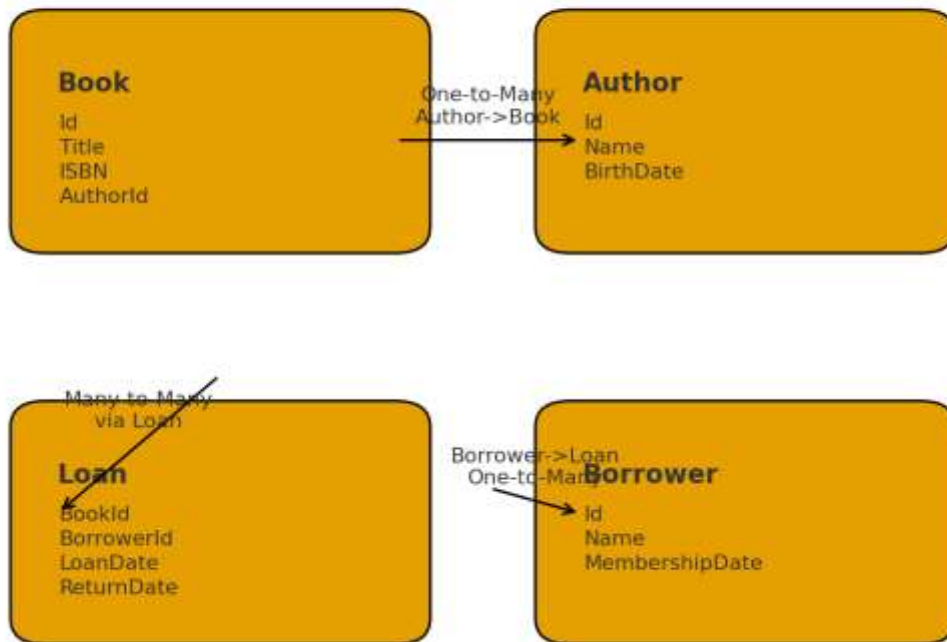


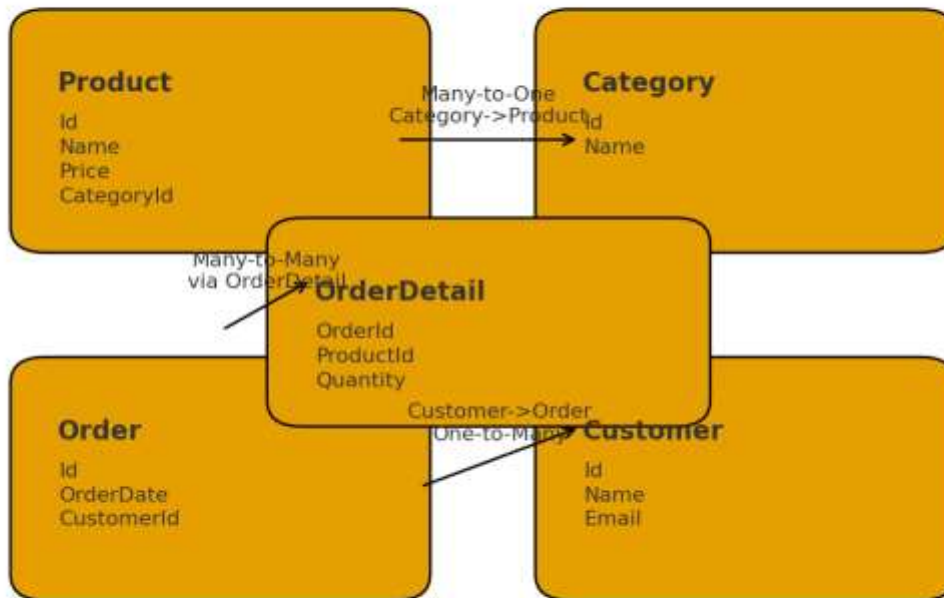
# Health Care System ER



# Library System ER



## E-Commerce System ER



---

```
using System;
```

```
using System.Collections.Generic;
```

```
using Microsoft.EntityFrameworkCore;
```

```
namespace ECommerce
```

```
{
```

```
    public class Product
```

```
    {
```

```
        public int Id { get; set; }
```

```
        public string Name { get; set; }
```

```
        public decimal Price { get; set; }
```

```
        public int CategoryId { get; set; }
```

```
        public Category Category { get; set; }
```

```
        public ICollection<OrderDetail> OrderDetails { get; set; }
```

```
}
```

```
public class Category
```

```
{
```

```
    public int Id { get; set; }
```

```
    public string Name { get; set; }
```

```
    public ICollection<Product> Products { get; set; }
```

```
}
```

```
public class Order
```

```
{
```

```
    public int Id { get; set; }
```

```
    public DateTime OrderDate { get; set; }
```

```
    public int CustomerId { get; set; }
```

```
    public Customer Customer { get; set; }
```

```
    public ICollection<OrderDetail> OrderDetails { get; set; }
```

```
}
```

```
public class Customer
```

```
{
```

```
    public int Id { get; set; }
```

```
    public string Name { get; set; }
```

```
    public string Email { get; set; }
```

```
    public ICollection<Order> Orders { get; set; }
```

```
}
```

```
public class OrderDetail
```

```
{
```

```
    public int OrderId { get; set; }
```

```
    public Order Order { get; set; }
```

```
    public int ProductId { get; set; }
```

```
public Product Product { get; set; }

public int Quantity { get; set; }

}
```

```
public class ECommerceContext : DbContext
{
    public DbSet<Product> Products { get; set; }
    public DbSet<Category> Categories { get; set; }
    public DbSet<Order> Orders { get; set; }
    public DbSet<Customer> Customers { get; set; }
    public DbSet<OrderDetail> OrderDetails { get; set; }

    protected override void OnModelCreating(ModelBuilder modelBuilder)
    {
        // Category - Product (One-to-Many)
        modelBuilder.Entity<Category>()
            .HasMany(c => c.Products)
            .WithOne(p => p.Category)
            .HasForeignKey(p => p.CategoryId)
            .OnDelete(DeleteBehavior.Cascade);

        // OrderDetail composite key
        modelBuilder.Entity<OrderDetail>()
            .HasKey(od => new { od.OrderId, od.ProductId });

        // Order - OrderDetail (One-to-Many)
        modelBuilder.Entity<Order>()
            .HasMany(o => o.OrderDetails)
            .WithOne(od => od.Order)
            .HasForeignKey(od => od.OrderId);
    }
}
```

```

// Product - OrderDetail (One-to-Many)

modelBuilder.Entity<Product>()
    .HasMany(p => p.OrderDetails)
    .WithOne(od => od.Product)
    .HasForeignKey(od => od.ProductId);

// Customer - Order (One-to-Many)

modelBuilder.Entity<Customer>()
    .HasMany(c => c.Orders)
    .WithOne(o => o.Customer)
    .HasForeignKey(o => o.CustomerId);
}
}
}

```

---

```
using System;
```

```
using System.Collections.Generic;
```

```
using Microsoft.EntityFrameworkCore;
```

```
namespace LibrarySystem
```

```
{
```

```
    public class Book
```

```
    {
```

```
        public int Id { get; set; }
```

```
        public string Title { get; set; }
```

```
        public string ISBN { get; set; }
```

```
        public int AuthorId { get; set; }
```

```
        public Author Author { get; set; }
```

```
        public ICollection<Loan> Loans { get; set; }
```

```
    }
```

```
    public class Author
```

```
{  
    public int Id { get; set; }  
    public string Name { get; set; }  
    public DateTime BirthDate { get; set; }  
    public ICollection<Book> Books { get; set; }  
}
```

```
public class Borrower  
{  
    public int Id { get; set; }  
    public string Name { get; set; }  
    public DateTime MembershipDate { get; set; }  
    public ICollection<Loan> Loans { get; set; }  
}
```

```
public class Loan  
{  
    public int BookId { get; set; }  
    public Book Book { get; set; }  
    public int BorrowerId { get; set; }  
    public Borrower Borrower { get; set; }  
    public DateTime LoanDate { get; set; }  
    public DateTime? ReturnDate { get; set; }  
}
```

```
public class LibraryContext : DbContext  
{  
    public DbSet<Book> Books { get; set; }  
    public DbSet<Author> Authors { get; set; }  
    public DbSet<Borrower> Borrowers { get; set; }  
    public DbSet<Loan> Loans { get; set; }
```

```

protected override void OnModelCreating(ModelBuilder modelBuilder)
{
    // Author - Book (One-to-Many)
    modelBuilder.Entity<Author>()
        .HasMany(a => a.Books)
        .WithOne(b => b.Author)
        .HasForeignKey(b => b.AuthorId);

    // Loan composite key
    modelBuilder.Entity<Loan>()
        .HasKey(l => new { l.BookId, l.BorrowerId, l.LoanDate });

    // Book - Loan (One-to-Many)
    modelBuilder.Entity<Book>()
        .HasMany(b => b.Loans)
        .WithOne(l => l.Book)
        .HasForeignKey(l => l.BookId);

    // Borrower - Loan (One-to-Many)
    modelBuilder.Entity<Borrower>()
        .HasMany(b => b.Loans)
        .WithOne(l => l.Borrower)
        .HasForeignKey(l => l.BorrowerId);
}
}
}

```

---

```

using System;
using System.Collections.Generic;
using Microsoft.EntityFrameworkCore;

```



```

namespace HealthCare
{
    public class Patient
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public DateTime DateOfBirth { get; set; }
        public ICollection<Appointment> Appointments { get; set; }
    }

    public class Doctor
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public string Specialization { get; set; }
        public ICollection<Appointment> Appointments { get; set; }
    }

    public class Appointment
    {
        public int PatientId { get; set; }
        public Patient Patient { get; set; }
        public int DoctorId { get; set; }
        public Doctor Doctor { get; set; }
        public DateTime AppointmentDate { get; set; }
    }

    public class HealthCareContext : DbContext
    {
        public DbSet<Patient> Patients { get; set; }
        public DbSet<Doctor> Doctors { get; set; }
    }
}

```

```

public DbSet<Appointment> Appointments { get; set; }

protected override void OnModelCreating(ModelBuilder modelBuilder)
{
    // Appointment composite key
    modelBuilder.Entity<Appointment>()
        .HasKey(a => new { a.PatientId, a.DoctorId, a.AppointmentDate });

    // Patient - Appointment (One-to-Many)
    modelBuilder.Entity<Patient>()
        .HasMany(p => p.Appointments)
        .WithOne(a => a.Patient)
        .HasForeignKey(a => a.PatientId);

    // Doctor - Appointment (One-to-Many)
    modelBuilder.Entity<Doctor>()
        .HasMany(d => d.Appointments)
        .WithOne(a => a.Doctor)
        .HasForeignKey(a => a.DoctorId);
}
}
}

```